

Mode 12 — Square-BFP (centered / opt)

C-DP4 (C16×C16) → DP8 (16×16) → DP8 (16×8) → DP8 (16×4) → DP8 (8×4) → block exponents → DP8 (4×4) → centered add-and-square

Step 1 — C-DP4 (C16×C16)

$$\text{C-DP4 (C16} \times \text{C16)} = D = \sum_{i=0}^3 a_i b_i, \quad a_i = a_i^{re} + j a_i^{im}, \quad b_i = b_i^{re} + j b_i^{im}$$

Step 2 — DP8 (16×16)

$$\text{Re}(D) = \underbrace{\sum_{i=0}^3 a_i^{re} b_i^{re} - \sum_{i=0}^3 a_i^{im} b_i^{im}}_{\text{DP8 (16} \times \text{16)}}$$

$$\text{Im}(D) = \underbrace{\sum_{i=0}^3 a_i^{re} b_i^{im} + \sum_{i=0}^3 a_i^{im} b_i^{re}}_{\text{DP8 (16} \times \text{16)}}$$

Step 3 — DP8 (16×8)

$$\text{Re}(D) = \underbrace{2^8 \sum_{i=0}^3 a_i^{re} b_i^{re,hi} - 2^8 \sum_{i=0}^3 a_i^{im} b_i^{im,hi}}_{\text{DP8 (16} \times \text{8)}} + \underbrace{\sum_{i=0}^3 a_i^{re} b_i^{re,lo} - \sum_{i=0}^3 a_i^{im} b_i^{im,lo}}_{\text{DP8 (16} \times \text{8)}}$$

$$\text{Im}(D) = \underbrace{2^8 \sum_{i=0}^3 a_i^{re} b_i^{im,hi} + 2^8 \sum_{i=0}^3 a_i^{im} b_i^{re,hi}}_{\text{DP8 (16} \times \text{8)}} + \underbrace{\sum_{i=0}^3 a_i^{re} b_i^{im,lo} + \sum_{i=0}^3 a_i^{im} b_i^{re,lo}}_{\text{DP8 (16} \times \text{8)}}$$

Step 4 — DP8 (16×4)

$$\begin{aligned} \text{Re}(D) = & \underbrace{2^{12} \sum_{i=0}^3 a_i^{re} b_i^{re,hh} - 2^{12} \sum_{i=0}^3 a_i^{im} b_i^{im,hh}}_{\text{DP8 (16} \times \text{4)}} \\ & + \underbrace{2^8 \sum_{i=0}^3 a_i^{re} b_i^{re,hl} - 2^8 \sum_{i=0}^3 a_i^{im} b_i^{im,hl}}_{\text{DP8 (16} \times \text{4)}} \\ & + \underbrace{2^4 \sum_{i=0}^3 a_i^{re} b_i^{re,lh} - 2^4 \sum_{i=0}^3 a_i^{im} b_i^{im,lh}}_{\text{DP8 (16} \times \text{4)}} \\ & + \underbrace{\sum_{i=0}^3 a_i^{re} b_i^{re,ll} - \sum_{i=0}^3 a_i^{im} b_i^{im,ll}}_{\text{DP8 (16} \times \text{4)}} \end{aligned}$$

$$\begin{aligned} \text{Im}(D) = & \underbrace{2^{12} \sum_{i=0}^3 a_i^{re} b_i^{im,hh} + 2^{12} \sum_{i=0}^3 a_i^{im} b_i^{re,hh}}_{\text{DP8 (16} \times \text{4)}} \\ & + \underbrace{2^8 \sum_{i=0}^3 a_i^{re} b_i^{im,hl} + 2^8 \sum_{i=0}^3 a_i^{im} b_i^{re,hl}}_{\text{DP8 (16} \times \text{4)}} \end{aligned}$$

$$\begin{aligned}
& + \underbrace{2^4 \sum_{i=0}^3 a_i^{re} b_i^{im, lh} + 2^4 \sum_{i=0}^3 a_i^{im} b_i^{re, lh}}_{\text{DP8 (16}\times\text{4)}} \\
& + \underbrace{\sum_{i=0}^3 a_i^{re} b_i^{im, ll} + \sum_{i=0}^3 a_i^{im} b_i^{re, ll}}_{\text{DP8 (16}\times\text{4)}}
\end{aligned}$$

Step 5 — DP8 (8×4)

$$\begin{aligned}
\text{Re}(D) = & \underbrace{2^{20} \sum_{i=0}^3 a_i^{re, hi} b_i^{re, hh}}_{\text{DP8 (8}\times\text{4)}} - \underbrace{2^{20} \sum_{i=0}^3 a_i^{im, hi} b_i^{im, hh}}_{\text{DP8 (8}\times\text{4)}} \\
& + \underbrace{2^{12} \sum_{i=0}^3 a_i^{re, lo} b_i^{re, hh}}_{\text{DP8 (8}\times\text{4)}} - \underbrace{2^{12} \sum_{i=0}^3 a_i^{im, lo} b_i^{im, hh}}_{\text{DP8 (8}\times\text{4)}} \\
& + \underbrace{2^{16} \sum_{i=0}^3 a_i^{re, hi} b_i^{re, hl}}_{\text{DP8 (8}\times\text{4)}} - \underbrace{2^{16} \sum_{i=0}^3 a_i^{im, hi} b_i^{im, hl}}_{\text{DP8 (8}\times\text{4)}} \\
& + \underbrace{2^8 \sum_{i=0}^3 a_i^{re, lo} b_i^{re, hl}}_{\text{DP8 (8}\times\text{4)}} - \underbrace{2^8 \sum_{i=0}^3 a_i^{im, lo} b_i^{im, hl}}_{\text{DP8 (8}\times\text{4)}} \\
& + \underbrace{2^{12} \sum_{i=0}^3 a_i^{re, hi} b_i^{re, lh}}_{\text{DP8 (8}\times\text{4)}} - \underbrace{2^{12} \sum_{i=0}^3 a_i^{im, hi} b_i^{im, lh}}_{\text{DP8 (8}\times\text{4)}} \\
& + \underbrace{2^4 \sum_{i=0}^3 a_i^{re, lo} b_i^{re, lh}}_{\text{DP8 (8}\times\text{4)}} - \underbrace{2^4 \sum_{i=0}^3 a_i^{im, lo} b_i^{im, lh}}_{\text{DP8 (8}\times\text{4)}} \\
& + \underbrace{2^8 \sum_{i=0}^3 a_i^{re, hi} b_i^{re, ll}}_{\text{DP8 (8}\times\text{4)}} - \underbrace{2^8 \sum_{i=0}^3 a_i^{im, hi} b_i^{im, ll}}_{\text{DP8 (8}\times\text{4)}} \\
& + \underbrace{\sum_{i=0}^3 a_i^{re, lo} b_i^{re, ll}}_{\text{DP8 (8}\times\text{4)}} - \underbrace{\sum_{i=0}^3 a_i^{im, lo} b_i^{im, ll}}_{\text{DP8 (8}\times\text{4)}}
\end{aligned}$$

$$\begin{aligned}
\text{Im}(D) = & \underbrace{2^{20} \sum_{i=0}^3 a_i^{re, hi} b_i^{im, hh}}_{\text{DP8 (8}\times\text{4)}} + \underbrace{2^{20} \sum_{i=0}^3 a_i^{im, hi} b_i^{re, hh}}_{\text{DP8 (8}\times\text{4)}} \\
& + \underbrace{2^{12} \sum_{i=0}^3 a_i^{re, lo} b_i^{im, hh}}_{\text{DP8 (8}\times\text{4)}} + \underbrace{2^{12} \sum_{i=0}^3 a_i^{im, lo} b_i^{re, hh}}_{\text{DP8 (8}\times\text{4)}} \\
& + \underbrace{2^{16} \sum_{i=0}^3 a_i^{re, hi} b_i^{im, hl}}_{\text{DP8 (8}\times\text{4)}} + \underbrace{2^{16} \sum_{i=0}^3 a_i^{im, hi} b_i^{re, hl}}_{\text{DP8 (8}\times\text{4)}}
\end{aligned}$$

$$\begin{aligned}
& + \underbrace{2^8 \sum_{i=0}^3 a_i^{re,lo} b_i^{im,hl}}_{\text{DP8 (8}\times\text{4)}} + \underbrace{2^8 \sum_{i=0}^3 a_i^{im,lo} b_i^{re,hl}}_{\text{DP8 (8}\times\text{4)}} \\
& + \underbrace{2^{12} \sum_{i=0}^3 a_i^{re,hi} b_i^{im,lh}}_{\text{DP8 (8}\times\text{4)}} + \underbrace{2^{12} \sum_{i=0}^3 a_i^{im,hi} b_i^{re,lh}}_{\text{DP8 (8}\times\text{4)}} \\
& + \underbrace{2^4 \sum_{i=0}^3 a_i^{re,lo} b_i^{im,lh}}_{\text{DP8 (8}\times\text{4)}} + \underbrace{2^4 \sum_{i=0}^3 a_i^{im,lo} b_i^{re,lh}}_{\text{DP8 (8}\times\text{4)}} \\
& + \underbrace{2^8 \sum_{i=0}^3 a_i^{re,hi} b_i^{im,ll}}_{\text{DP8 (8}\times\text{4)}} + \underbrace{2^8 \sum_{i=0}^3 a_i^{im,hi} b_i^{re,ll}}_{\text{DP8 (8}\times\text{4)}} \\
& + \underbrace{\sum_{i=0}^3 a_i^{re,lo} b_i^{im,ll}}_{\text{DP8 (8}\times\text{4)}} + \underbrace{\sum_{i=0}^3 a_i^{im,lo} b_i^{re,ll}}_{\text{DP8 (8}\times\text{4)}}
\end{aligned}$$

Step 6 — Block exponents (one exponent block per DP8 (8×4) — all copies of one E ; re/im and both B arrangements tied, negation scale-blind)

$$(P_j, E) \text{ per DP8 (8}\times\text{4) above,} \quad E = E_a + E_b \longrightarrow (M^{re}, E), \quad (M^{im}, E)$$

$\delta \equiv 0$: M^{re} , M^{im} are the Step-5 sums unchanged; $\text{Re}(\hat{D}) = M^{re} \cdot 2^E$, $\text{Im}(\hat{D}) = M^{im} \cdot 2^E$.

Step 7 — DP8 (4×4)

$$\begin{aligned}
M^{re} = & \underbrace{2^{24} \sum_{i=0}^3 a_{H,i}^{re,hi} b_i^{re,hh}}_{\text{DP8 (4}\times\text{4)}} + \underbrace{2^{20} \sum_{i=0}^3 a_{L,i}^{re,hi} b_i^{re,hh}}_{\text{DP8 (4}\times\text{4)}} \\
& - \underbrace{2^{24} \sum_{i=0}^3 a_{H,i}^{im,hi} b_i^{im,hh}}_{\text{DP8 (4}\times\text{4)}} - \underbrace{2^{20} \sum_{i=0}^3 a_{L,i}^{im,hi} b_i^{im,hh}}_{\text{DP8 (4}\times\text{4)}} \\
& + \underbrace{2^{16} \sum_{i=0}^3 a_{H,i}^{re,lo} b_i^{re,hh}}_{\text{DP8 (4}\times\text{4)}} + \underbrace{2^{12} \sum_{i=0}^3 a_{L,i}^{re,lo} b_i^{re,hh}}_{\text{DP8 (4}\times\text{4)}} \\
& - \underbrace{2^{16} \sum_{i=0}^3 a_{H,i}^{im,lo} b_i^{im,hh}}_{\text{DP8 (4}\times\text{4)}} - \underbrace{2^{12} \sum_{i=0}^3 a_{L,i}^{im,lo} b_i^{im,hh}}_{\text{DP8 (4}\times\text{4)}} \\
& + \underbrace{2^{20} \sum_{i=0}^3 a_{H,i}^{re,hi} b_i^{re,hl}}_{\text{DP8 (4}\times\text{4)}} + \underbrace{2^{16} \sum_{i=0}^3 a_{L,i}^{re,hi} b_i^{re,hl}}_{\text{DP8 (4}\times\text{4)}} \\
& - \underbrace{2^{20} \sum_{i=0}^3 a_{H,i}^{im,hi} b_i^{im,hl}}_{\text{DP8 (4}\times\text{4)}} - \underbrace{2^{16} \sum_{i=0}^3 a_{L,i}^{im,hi} b_i^{im,hl}}_{\text{DP8 (4}\times\text{4)}} \\
& + \underbrace{2^{12} \sum_{i=0}^3 a_{H,i}^{re,lo} b_i^{re,hl}}_{\text{DP8 (4}\times\text{4)}} + \underbrace{2^8 \sum_{i=0}^3 a_{L,i}^{re,lo} b_i^{re,hl}}_{\text{DP8 (4}\times\text{4)}}
\end{aligned}$$

$$\begin{aligned}
& -2^{12} \underbrace{\sum_{i=0}^3 a_{H,i}^{im,lo} b_i^{im,hl}}_{\text{DP8 (4x4)}} - 2^8 \underbrace{\sum_{i=0}^3 a_{L,i}^{im,lo} b_i^{im,hl}}_{\text{DP8 (4x4)}} \\
& + 2^{16} \underbrace{\sum_{i=0}^3 a_{H,i}^{re,hi} b_i^{re,lh}}_{\text{DP8 (4x4)}} + 2^{12} \underbrace{\sum_{i=0}^3 a_{L,i}^{re,hi} b_i^{re,lh}}_{\text{DP8 (4x4)}} \\
& - 2^{16} \underbrace{\sum_{i=0}^3 a_{H,i}^{im,hi} b_i^{im,lh}}_{\text{DP8 (4x4)}} - 2^{12} \underbrace{\sum_{i=0}^3 a_{L,i}^{im,hi} b_i^{im,lh}}_{\text{DP8 (4x4)}} \\
& + 2^8 \underbrace{\sum_{i=0}^3 a_{H,i}^{re,lo} b_i^{re,lh}}_{\text{DP8 (4x4)}} + 2^4 \underbrace{\sum_{i=0}^3 a_{L,i}^{re,lo} b_i^{re,lh}}_{\text{DP8 (4x4)}} \\
& - 2^8 \underbrace{\sum_{i=0}^3 a_{H,i}^{im,lo} b_i^{im,lh}}_{\text{DP8 (4x4)}} - 2^4 \underbrace{\sum_{i=0}^3 a_{L,i}^{im,lo} b_i^{im,lh}}_{\text{DP8 (4x4)}} \\
& + 2^{12} \underbrace{\sum_{i=0}^3 a_{H,i}^{re,hi} b_i^{re,ll}}_{\text{DP8 (4x4)}} + 2^8 \underbrace{\sum_{i=0}^3 a_{L,i}^{re,hi} b_i^{re,ll}}_{\text{DP8 (4x4)}} \\
& - 2^{12} \underbrace{\sum_{i=0}^3 a_{H,i}^{im,hi} b_i^{im,ll}}_{\text{DP8 (4x4)}} - 2^8 \underbrace{\sum_{i=0}^3 a_{L,i}^{im,hi} b_i^{im,ll}}_{\text{DP8 (4x4)}} \\
& + 2^4 \underbrace{\sum_{i=0}^3 a_{H,i}^{re,lo} b_i^{re,ll}}_{\text{DP8 (4x4)}} + \underbrace{\sum_{i=0}^3 a_{L,i}^{re,lo} b_i^{re,ll}}_{\text{DP8 (4x4)}} \\
& - 2^4 \underbrace{\sum_{i=0}^3 a_{H,i}^{im,lo} b_i^{im,ll}}_{\text{DP8 (4x4)}} - \underbrace{\sum_{i=0}^3 a_{L,i}^{im,lo} b_i^{im,ll}}_{\text{DP8 (4x4)}}
\end{aligned}$$

$$\begin{aligned}
M^{im} = & 2^{24} \underbrace{\sum_{i=0}^3 a_{H,i}^{re,hi} b_i^{im,hh}}_{\text{DP8 (4x4)}} + 2^{20} \underbrace{\sum_{i=0}^3 a_{L,i}^{re,hi} b_i^{im,hh}}_{\text{DP8 (4x4)}} \\
& + 2^{24} \underbrace{\sum_{i=0}^3 a_{H,i}^{im,hi} b_i^{re,hh}}_{\text{DP8 (4x4)}} + 2^{20} \underbrace{\sum_{i=0}^3 a_{L,i}^{im,hi} b_i^{re,hh}}_{\text{DP8 (4x4)}} \\
& + 2^{16} \underbrace{\sum_{i=0}^3 a_{H,i}^{re,lo} b_i^{im,hh}}_{\text{DP8 (4x4)}} + 2^{12} \underbrace{\sum_{i=0}^3 a_{L,i}^{re,lo} b_i^{im,hh}}_{\text{DP8 (4x4)}} \\
& + 2^{16} \underbrace{\sum_{i=0}^3 a_{H,i}^{im,lo} b_i^{re,hh}}_{\text{DP8 (4x4)}} + 2^{12} \underbrace{\sum_{i=0}^3 a_{L,i}^{im,lo} b_i^{re,hh}}_{\text{DP8 (4x4)}}
\end{aligned}$$

$$\begin{aligned}
& + 2^{20} \underbrace{\sum_{i=0}^3 a_{H,i}^{re,hi} b_i^{im,hl}}_{\text{DP8 (4x4)}} + 2^{16} \underbrace{\sum_{i=0}^3 a_{L,i}^{re,hi} b_i^{im,hl}}_{\text{DP8 (4x4)}} \\
& + 2^{20} \underbrace{\sum_{i=0}^3 a_{H,i}^{im,hi} b_i^{re,hl}}_{\text{DP8 (4x4)}} + 2^{16} \underbrace{\sum_{i=0}^3 a_{L,i}^{im,hi} b_i^{re,hl}}_{\text{DP8 (4x4)}} \\
& + 2^{12} \underbrace{\sum_{i=0}^3 a_{H,i}^{re,lo} b_i^{im,hl}}_{\text{DP8 (4x4)}} + 2^8 \underbrace{\sum_{i=0}^3 a_{L,i}^{re,lo} b_i^{im,hl}}_{\text{DP8 (4x4)}} \\
& + 2^{12} \underbrace{\sum_{i=0}^3 a_{H,i}^{im,lo} b_i^{re,hl}}_{\text{DP8 (4x4)}} + 2^8 \underbrace{\sum_{i=0}^3 a_{L,i}^{im,lo} b_i^{re,hl}}_{\text{DP8 (4x4)}} \\
& + 2^{16} \underbrace{\sum_{i=0}^3 a_{H,i}^{re,hi} b_i^{im,lh}}_{\text{DP8 (4x4)}} + 2^{12} \underbrace{\sum_{i=0}^3 a_{L,i}^{re,hi} b_i^{im,lh}}_{\text{DP8 (4x4)}} \\
& + 2^{16} \underbrace{\sum_{i=0}^3 a_{H,i}^{im,hi} b_i^{re,lh}}_{\text{DP8 (4x4)}} + 2^{12} \underbrace{\sum_{i=0}^3 a_{L,i}^{im,hi} b_i^{re,lh}}_{\text{DP8 (4x4)}} \\
& + 2^8 \underbrace{\sum_{i=0}^3 a_{H,i}^{re,lo} b_i^{im,lh}}_{\text{DP8 (4x4)}} + 2^4 \underbrace{\sum_{i=0}^3 a_{L,i}^{re,lo} b_i^{im,lh}}_{\text{DP8 (4x4)}} \\
& + 2^8 \underbrace{\sum_{i=0}^3 a_{H,i}^{im,lo} b_i^{re,lh}}_{\text{DP8 (4x4)}} + 2^4 \underbrace{\sum_{i=0}^3 a_{L,i}^{im,lo} b_i^{re,lh}}_{\text{DP8 (4x4)}} \\
& + 2^{12} \underbrace{\sum_{i=0}^3 a_{H,i}^{re,hi} b_i^{im,ll}}_{\text{DP8 (4x4)}} + 2^8 \underbrace{\sum_{i=0}^3 a_{L,i}^{re,hi} b_i^{im,ll}}_{\text{DP8 (4x4)}} \\
& + 2^{12} \underbrace{\sum_{i=0}^3 a_{H,i}^{im,hi} b_i^{re,ll}}_{\text{DP8 (4x4)}} + 2^8 \underbrace{\sum_{i=0}^3 a_{L,i}^{im,hi} b_i^{re,ll}}_{\text{DP8 (4x4)}} \\
& + 2^4 \underbrace{\sum_{i=0}^3 a_{H,i}^{re,lo} b_i^{im,ll}}_{\text{DP8 (4x4)}} + \underbrace{\sum_{i=0}^3 a_{L,i}^{re,lo} b_i^{im,ll}}_{\text{DP8 (4x4)}} \\
& + 2^4 \underbrace{\sum_{i=0}^3 a_{H,i}^{im,lo} b_i^{re,ll}}_{\text{DP8 (4x4)}} + \underbrace{\sum_{i=0}^3 a_{L,i}^{im,lo} b_i^{re,ll}}_{\text{DP8 (4x4)}}
\end{aligned}$$

Step 8 — Centered add-and-square (subtract 8 from unsigned nibbles; per DP8 (4x4) add $c = 0/256/1024$ for 0/1/2 unsigned nibbles; negated $a^{im}b^{im}$ blocks carry the minus outside)

$$M^{re} = 2^{24} \underbrace{\frac{1}{2} \left(\sum_{i=0}^3 (a_{H,i}^{re,hi} + b_i^{re,hh})^2 - \sum_{i=0}^3 (a_{H,i}^{re,hi})^2 - \sum_{i=0}^3 (b_i^{re,hh})^2 \right)}_{\text{DP8 (4x4)}}$$

$$\begin{aligned}
& + 2^{20} \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 ((a_{L,i}^{re,hi} - 8) + b_i^{re,hh})^2 - \sum_{i=0}^3 (a_{L,i}^{re,hi} - 8)^2 - \sum_{i=0}^3 (b_i^{re,hh} - 8)^2 + 256 \right)}_{\text{DP8 (4x4)}} \\
& - 2^{24} \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{H,i}^{im,hi} + b_i^{im,hh})^2 - \sum_{i=0}^3 (a_{H,i}^{im,hi})^2 - \sum_{i=0}^3 (b_i^{im,hh})^2 \right)}_{\text{DP8 (4x4)}} \\
& - 2^{20} \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 ((a_{L,i}^{im,hi} - 8) + b_i^{im,hh})^2 - \sum_{i=0}^3 (a_{L,i}^{im,hi} - 8)^2 - \sum_{i=0}^3 (b_i^{im,hh} - 8)^2 + 256 \right)}_{\text{DP8 (4x4)}} \\
& + 2^{16} \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 ((a_{H,i}^{re,lo} - 8) + b_i^{re,hh})^2 - \sum_{i=0}^3 (a_{H,i}^{re,lo} - 8)^2 - \sum_{i=0}^3 (b_i^{re,hh} - 8)^2 + 256 \right)}_{\text{DP8 (4x4)}} \\
& + 2^{12} \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 ((a_{L,i}^{re,lo} - 8) + b_i^{re,hh})^2 - \sum_{i=0}^3 (a_{L,i}^{re,lo} - 8)^2 - \sum_{i=0}^3 (b_i^{re,hh} - 8)^2 + 256 \right)}_{\text{DP8 (4x4)}} \\
& - 2^{16} \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 ((a_{H,i}^{im,lo} - 8) + b_i^{im,hh})^2 - \sum_{i=0}^3 (a_{H,i}^{im,lo} - 8)^2 - \sum_{i=0}^3 (b_i^{im,hh} - 8)^2 + 256 \right)}_{\text{DP8 (4x4)}} \\
& - 2^{12} \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 ((a_{L,i}^{im,lo} - 8) + b_i^{im,hh})^2 - \sum_{i=0}^3 (a_{L,i}^{im,lo} - 8)^2 - \sum_{i=0}^3 (b_i^{im,hh} - 8)^2 + 256 \right)}_{\text{DP8 (4x4)}} \\
& + 2^{20} \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{H,i}^{re,hi} + (b_i^{re,hl} - 8))^2 - \sum_{i=0}^3 (a_{H,i}^{re,hi} - 8)^2 - \sum_{i=0}^3 (b_i^{re,hl} - 8)^2 + 256 \right)}_{\text{DP8 (4x4)}} \\
& + 2^{16} \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 ((a_{L,i}^{re,hi} - 8) + (b_i^{re,hl} - 8))^2 - \sum_{i=0}^3 (a_{L,i}^{re,hi} - 16)^2 - \sum_{i=0}^3 (b_i^{re,hl} - 16)^2 + 1024 \right)}_{\text{DP8 (4x4)}} \\
& - 2^{20} \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{H,i}^{im,hi} + (b_i^{im,hl} - 8))^2 - \sum_{i=0}^3 (a_{H,i}^{im,hi} - 8)^2 - \sum_{i=0}^3 (b_i^{im,hl} - 8)^2 + 256 \right)}_{\text{DP8 (4x4)}} \\
& - 2^{16} \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 ((a_{L,i}^{im,hi} - 8) + (b_i^{im,hl} - 8))^2 - \sum_{i=0}^3 (a_{L,i}^{im,hi} - 16)^2 - \sum_{i=0}^3 (b_i^{im,hl} - 16)^2 + 1024 \right)}_{\text{DP8 (4x4)}} \\
& + 2^{12} \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 ((a_{H,i}^{re,lo} - 8) + (b_i^{re,hl} - 8))^2 - \sum_{i=0}^3 (a_{H,i}^{re,lo} - 16)^2 - \sum_{i=0}^3 (b_i^{re,hl} - 16)^2 + 1024 \right)}_{\text{DP8 (4x4)}} \\
& + 2^8 \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 ((a_{L,i}^{re,lo} - 8) + (b_i^{re,hl} - 8))^2 - \sum_{i=0}^3 (a_{L,i}^{re,lo} - 16)^2 - \sum_{i=0}^3 (b_i^{re,hl} - 16)^2 + 1024 \right)}_{\text{DP8 (4x4)}} \\
& - 2^{12} \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 ((a_{H,i}^{im,lo} - 8) + (b_i^{im,hl} - 8))^2 - \sum_{i=0}^3 (a_{H,i}^{im,lo} - 16)^2 - \sum_{i=0}^3 (b_i^{im,hl} - 16)^2 + 1024 \right)}_{\text{DP8 (4x4)}}
\end{aligned}$$

$$\begin{aligned}
& - 2^8 \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 ((a_{L,i}^{im,lo} - 8) + (b_i^{im,hl} - 8))^2 - \sum_{i=0}^3 (a_{L,i}^{im,lo} - 16)^2 - \sum_{i=0}^3 (b_i^{im,hl} - 16)^2 + 1024 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^{16} \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{H,i}^{re,hi} + (b_i^{re,hl} - 8))^2 - \sum_{i=0}^3 (a_{H,i}^{re,hi} - 8)^2 - \sum_{i=0}^3 (b_i^{re,hl} - 8)^2 + 256 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^{12} \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 ((a_{L,i}^{re,hi} - 8) + (b_i^{re,hl} - 8))^2 - \sum_{i=0}^3 (a_{L,i}^{re,hi} - 16)^2 - \sum_{i=0}^3 (b_i^{re,hl} - 16)^2 + 1024 \right)}_{\text{DP8 } (4 \times 4)} \\
& - 2^{16} \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{H,i}^{im,hi} + (b_i^{im,hl} - 8))^2 - \sum_{i=0}^3 (a_{H,i}^{im,hi} - 8)^2 - \sum_{i=0}^3 (b_i^{im,hl} - 8)^2 + 256 \right)}_{\text{DP8 } (4 \times 4)} \\
& - 2^{12} \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 ((a_{L,i}^{im,hi} - 8) + (b_i^{im,hl} - 8))^2 - \sum_{i=0}^3 (a_{L,i}^{im,hi} - 16)^2 - \sum_{i=0}^3 (b_i^{im,hl} - 16)^2 + 1024 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^8 \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 ((a_{H,i}^{re,lo} - 8) + (b_i^{re,hl} - 8))^2 - \sum_{i=0}^3 (a_{H,i}^{re,lo} - 16)^2 - \sum_{i=0}^3 (b_i^{re,hl} - 16)^2 + 1024 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^4 \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 ((a_{L,i}^{re,lo} - 8) + (b_i^{re,hl} - 8))^2 - \sum_{i=0}^3 (a_{L,i}^{re,lo} - 16)^2 - \sum_{i=0}^3 (b_i^{re,hl} - 16)^2 + 1024 \right)}_{\text{DP8 } (4 \times 4)} \\
& - 2^8 \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 ((a_{H,i}^{im,lo} - 8) + (b_i^{im,hl} - 8))^2 - \sum_{i=0}^3 (a_{H,i}^{im,lo} - 16)^2 - \sum_{i=0}^3 (b_i^{im,hl} - 16)^2 + 1024 \right)}_{\text{DP8 } (4 \times 4)} \\
& - 2^4 \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 ((a_{L,i}^{im,lo} - 8) + (b_i^{im,hl} - 8))^2 - \sum_{i=0}^3 (a_{L,i}^{im,lo} - 16)^2 - \sum_{i=0}^3 (b_i^{im,hl} - 16)^2 + 1024 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^{12} \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{H,i}^{re,hi} + (b_i^{re,ll} - 8))^2 - \sum_{i=0}^3 (a_{H,i}^{re,hi} - 8)^2 - \sum_{i=0}^3 (b_i^{re,ll} - 8)^2 + 256 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^8 \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 ((a_{L,i}^{re,hi} - 8) + (b_i^{re,ll} - 8))^2 - \sum_{i=0}^3 (a_{L,i}^{re,hi} - 16)^2 - \sum_{i=0}^3 (b_i^{re,ll} - 16)^2 + 1024 \right)}_{\text{DP8 } (4 \times 4)} \\
& - 2^{12} \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{H,i}^{im,hi} + (b_i^{im,ll} - 8))^2 - \sum_{i=0}^3 (a_{H,i}^{im,hi} - 8)^2 - \sum_{i=0}^3 (b_i^{im,ll} - 8)^2 + 256 \right)}_{\text{DP8 } (4 \times 4)} \\
& - 2^8 \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 ((a_{L,i}^{im,hi} - 8) + (b_i^{im,ll} - 8))^2 - \sum_{i=0}^3 (a_{L,i}^{im,hi} - 16)^2 - \sum_{i=0}^3 (b_i^{im,ll} - 16)^2 + 1024 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^4 \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 ((a_{H,i}^{re,lo} - 8) + (b_i^{re,ll} - 8))^2 - \sum_{i=0}^3 (a_{H,i}^{re,lo} - 16)^2 - \sum_{i=0}^3 (b_i^{re,ll} - 16)^2 + 1024 \right)}_{\text{DP8 } (4 \times 4)}
\end{aligned}$$

$$\begin{aligned}
& + \underbrace{\frac{1}{2} \left(\sum_{i=0}^3 ((a_{L,i}^{re,lo} - 8) + (b_i^{re,ll} - 8))^2 - \sum_{i=0}^3 (a_{L,i}^{re,lo} - 16)^2 - \sum_{i=0}^3 (b_i^{re,ll} - 16)^2 + 1024 \right)}_{\text{DP8 (4x4)}} \\
& - \underbrace{2^4 \frac{1}{2} \left(\sum_{i=0}^3 ((a_{H,i}^{im,lo} - 8) + (b_i^{im,ll} - 8))^2 - \sum_{i=0}^3 (a_{H,i}^{im,lo} - 16)^2 - \sum_{i=0}^3 (b_i^{im,ll} - 16)^2 + 1024 \right)}_{\text{DP8 (4x4)}} \\
& - \underbrace{\frac{1}{2} \left(\sum_{i=0}^3 ((a_{L,i}^{im,lo} - 8) + (b_i^{im,ll} - 8))^2 - \sum_{i=0}^3 (a_{L,i}^{im,lo} - 16)^2 - \sum_{i=0}^3 (b_i^{im,ll} - 16)^2 + 1024 \right)}_{\text{DP8 (4x4)}} \\
M^{im} = & \underbrace{2^{24} \frac{1}{2} \left(\sum_{i=0}^3 (a_{H,i}^{re,hi} + b_i^{im,hh})^2 - \sum_{i=0}^3 (a_{H,i}^{re,hi})^2 - \sum_{i=0}^3 (b_i^{im,hh})^2 \right)}_{\text{DP8 (4x4)}} \\
& + \underbrace{2^{20} \frac{1}{2} \left(\sum_{i=0}^3 ((a_{L,i}^{re,hi} - 8) + b_i^{im,hh})^2 - \sum_{i=0}^3 (a_{L,i}^{re,hi} - 8)^2 - \sum_{i=0}^3 (b_i^{im,hh} - 8)^2 + 256 \right)}_{\text{DP8 (4x4)}} \\
& + \underbrace{2^{24} \frac{1}{2} \left(\sum_{i=0}^3 (a_{H,i}^{im,hi} + b_i^{re,hh})^2 - \sum_{i=0}^3 (a_{H,i}^{im,hi})^2 - \sum_{i=0}^3 (b_i^{re,hh})^2 \right)}_{\text{DP8 (4x4)}} \\
& + \underbrace{2^{20} \frac{1}{2} \left(\sum_{i=0}^3 ((a_{L,i}^{im,hi} - 8) + b_i^{re,hh})^2 - \sum_{i=0}^3 (a_{L,i}^{im,hi} - 8)^2 - \sum_{i=0}^3 (b_i^{re,hh} - 8)^2 + 256 \right)}_{\text{DP8 (4x4)}} \\
& + \underbrace{2^{16} \frac{1}{2} \left(\sum_{i=0}^3 ((a_{H,i}^{re,lo} - 8) + b_i^{im,hh})^2 - \sum_{i=0}^3 (a_{H,i}^{re,lo} - 8)^2 - \sum_{i=0}^3 (b_i^{im,hh} - 8)^2 + 256 \right)}_{\text{DP8 (4x4)}} \\
& + \underbrace{2^{12} \frac{1}{2} \left(\sum_{i=0}^3 ((a_{L,i}^{re,lo} - 8) + b_i^{im,hh})^2 - \sum_{i=0}^3 (a_{L,i}^{re,lo} - 8)^2 - \sum_{i=0}^3 (b_i^{im,hh} - 8)^2 + 256 \right)}_{\text{DP8 (4x4)}} \\
& + \underbrace{2^{16} \frac{1}{2} \left(\sum_{i=0}^3 ((a_{H,i}^{im,lo} - 8) + b_i^{re,hh})^2 - \sum_{i=0}^3 (a_{H,i}^{im,lo} - 8)^2 - \sum_{i=0}^3 (b_i^{re,hh} - 8)^2 + 256 \right)}_{\text{DP8 (4x4)}} \\
& + \underbrace{2^{12} \frac{1}{2} \left(\sum_{i=0}^3 ((a_{L,i}^{im,lo} - 8) + b_i^{re,hh})^2 - \sum_{i=0}^3 (a_{L,i}^{im,lo} - 8)^2 - \sum_{i=0}^3 (b_i^{re,hh} - 8)^2 + 256 \right)}_{\text{DP8 (4x4)}} \\
& + \underbrace{2^{20} \frac{1}{2} \left(\sum_{i=0}^3 (a_{H,i}^{re,hi} + (b_i^{im,hl} - 8))^2 - \sum_{i=0}^3 (a_{H,i}^{re,hi} - 8)^2 - \sum_{i=0}^3 (b_i^{im,hl} - 8)^2 + 256 \right)}_{\text{DP8 (4x4)}} \\
& + \underbrace{2^{16} \frac{1}{2} \left(\sum_{i=0}^3 ((a_{L,i}^{re,hi} - 8) + (b_i^{im,hl} - 8))^2 - \sum_{i=0}^3 (a_{L,i}^{re,hi} - 16)^2 - \sum_{i=0}^3 (b_i^{im,hl} - 16)^2 + 1024 \right)}_{\text{DP8 (4x4)}} \\
& + \underbrace{2^{20} \frac{1}{2} \left(\sum_{i=0}^3 (a_{H,i}^{im,hi} + (b_i^{re,hl} - 8))^2 - \sum_{i=0}^3 (a_{H,i}^{im,hi} - 8)^2 - \sum_{i=0}^3 (b_i^{re,hl} - 8)^2 + 256 \right)}_{\text{DP8 (4x4)}}
\end{aligned}$$

$$\begin{aligned}
& + 2^{16} \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 ((a_{L,i}^{im,hi} - 8) + (b_i^{re,hl} - 8))^2 - \sum_{i=0}^3 (a_{L,i}^{im,hi} - 16)^2 - \sum_{i=0}^3 (b_i^{re,hl} - 16)^2 + 1024 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^{12} \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 ((a_{H,i}^{re,lo} - 8) + (b_i^{im,hl} - 8))^2 - \sum_{i=0}^3 (a_{H,i}^{re,lo} - 16)^2 - \sum_{i=0}^3 (b_i^{im,hl} - 16)^2 + 1024 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^8 \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 ((a_{L,i}^{re,lo} - 8) + (b_i^{im,hl} - 8))^2 - \sum_{i=0}^3 (a_{L,i}^{re,lo} - 16)^2 - \sum_{i=0}^3 (b_i^{im,hl} - 16)^2 + 1024 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^{12} \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 ((a_{H,i}^{im,lo} - 8) + (b_i^{re,hl} - 8))^2 - \sum_{i=0}^3 (a_{H,i}^{im,lo} - 16)^2 - \sum_{i=0}^3 (b_i^{re,hl} - 16)^2 + 1024 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^8 \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 ((a_{L,i}^{im,lo} - 8) + (b_i^{re,hl} - 8))^2 - \sum_{i=0}^3 (a_{L,i}^{im,lo} - 16)^2 - \sum_{i=0}^3 (b_i^{re,hl} - 16)^2 + 1024 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^{16} \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{H,i}^{re,hi} + (b_i^{im,lh} - 8))^2 - \sum_{i=0}^3 (a_{H,i}^{re,hi} - 8)^2 - \sum_{i=0}^3 (b_i^{im,lh} - 8)^2 + 256 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^{12} \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 ((a_{L,i}^{re,hi} - 8) + (b_i^{im,lh} - 8))^2 - \sum_{i=0}^3 (a_{L,i}^{re,hi} - 16)^2 - \sum_{i=0}^3 (b_i^{im,lh} - 16)^2 + 1024 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^{16} \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{H,i}^{im,hi} + (b_i^{re,lh} - 8))^2 - \sum_{i=0}^3 (a_{H,i}^{im,hi} - 8)^2 - \sum_{i=0}^3 (b_i^{re,lh} - 8)^2 + 256 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^{12} \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 ((a_{L,i}^{im,hi} - 8) + (b_i^{re,lh} - 8))^2 - \sum_{i=0}^3 (a_{L,i}^{im,hi} - 16)^2 - \sum_{i=0}^3 (b_i^{re,lh} - 16)^2 + 1024 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^8 \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 ((a_{H,i}^{re,lo} - 8) + (b_i^{im,lh} - 8))^2 - \sum_{i=0}^3 (a_{H,i}^{re,lo} - 16)^2 - \sum_{i=0}^3 (b_i^{im,lh} - 16)^2 + 1024 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^4 \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 ((a_{L,i}^{re,lo} - 8) + (b_i^{im,lh} - 8))^2 - \sum_{i=0}^3 (a_{L,i}^{re,lo} - 16)^2 - \sum_{i=0}^3 (b_i^{im,lh} - 16)^2 + 1024 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^8 \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 ((a_{H,i}^{im,lo} - 8) + (b_i^{re,lh} - 8))^2 - \sum_{i=0}^3 (a_{H,i}^{im,lo} - 16)^2 - \sum_{i=0}^3 (b_i^{re,lh} - 16)^2 + 1024 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^4 \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 ((a_{L,i}^{im,lo} - 8) + (b_i^{re,lh} - 8))^2 - \sum_{i=0}^3 (a_{L,i}^{im,lo} - 16)^2 - \sum_{i=0}^3 (b_i^{re,lh} - 16)^2 + 1024 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^{12} \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 (a_{H,i}^{re,hi} + (b_i^{im,ll} - 8))^2 - \sum_{i=0}^3 (a_{H,i}^{re,hi} - 8)^2 - \sum_{i=0}^3 (b_i^{im,ll} - 8)^2 + 256 \right)}_{\text{DP8 } (4 \times 4)}
\end{aligned}$$

$$\begin{aligned}
& + 2^8 \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 \left((a_{L,i}^{re,hi} - 8) + (b_i^{im,ll} - 8) \right)^2 - \sum_{i=0}^3 (a_{L,i}^{re,hi} - 16)^2 - \sum_{i=0}^3 (b_i^{im,ll} - 16)^2 + 1024 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^{12} \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 \left((a_{H,i}^{im,hi} + (b_i^{re,ll} - 8)) \right)^2 - \sum_{i=0}^3 (a_{H,i}^{im,hi} - 8)^2 - \sum_{i=0}^3 (b_i^{re,ll} - 8)^2 + 256 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^8 \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 \left((a_{L,i}^{im,hi} - 8) + (b_i^{re,ll} - 8) \right)^2 - \sum_{i=0}^3 (a_{L,i}^{im,hi} - 16)^2 - \sum_{i=0}^3 (b_i^{re,ll} - 16)^2 + 1024 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^4 \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 \left((a_{H,i}^{re,lo} - 8) + (b_i^{im,ll} - 8) \right)^2 - \sum_{i=0}^3 (a_{H,i}^{re,lo} - 16)^2 - \sum_{i=0}^3 (b_i^{im,ll} - 16)^2 + 1024 \right)}_{\text{DP8 } (4 \times 4)} \\
& + \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 \left((a_{L,i}^{re,lo} - 8) + (b_i^{im,ll} - 8) \right)^2 - \sum_{i=0}^3 (a_{L,i}^{re,lo} - 16)^2 - \sum_{i=0}^3 (b_i^{im,ll} - 16)^2 + 1024 \right)}_{\text{DP8 } (4 \times 4)} \\
& + 2^4 \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 \left((a_{H,i}^{im,lo} - 8) + (b_i^{re,ll} - 8) \right)^2 - \sum_{i=0}^3 (a_{H,i}^{im,lo} - 16)^2 - \sum_{i=0}^3 (b_i^{re,ll} - 16)^2 + 1024 \right)}_{\text{DP8 } (4 \times 4)} \\
& + \frac{1}{2} \underbrace{\left(\sum_{i=0}^3 \left((a_{L,i}^{im,lo} - 8) + (b_i^{re,ll} - 8) \right)^2 - \sum_{i=0}^3 (a_{L,i}^{im,lo} - 16)^2 - \sum_{i=0}^3 (b_i^{re,ll} - 16)^2 + 1024 \right)}_{\text{DP8 } (4 \times 4)}
\end{aligned}$$

Step 9 — All components

Re (M^{re}):

$$\begin{aligned}
PE &= 2^{24} \sum_{i=0}^3 (a_{H,i}^{re,hi} + b_i^{re,hh})^2 + 2^{20} \sum_{i=0}^3 ((a_{L,i}^{re,hi} - 8) + b_i^{re,hh})^2 - 2^{24} \sum_{i=0}^3 (a_{H,i}^{im,hi} + b_i^{im,hh})^2 - 2^{20} \sum_{i=0}^3 ((a_{L,i}^{im,hi} - 8) + b_i^{im,hh})^2 \\
&+ 2^{16} \sum_{i=0}^3 ((a_{H,i}^{re,lo} - 8) + b_i^{re,hh})^2 + 2^{12} \sum_{i=0}^3 ((a_{L,i}^{re,lo} - 8) + b_i^{re,hh})^2 - 2^{16} \sum_{i=0}^3 ((a_{H,i}^{im,lo} - 8) + b_i^{im,hh})^2 - 2^{12} \sum_{i=0}^3 ((a_{L,i}^{im,lo} - 8) + b_i^{im,hh})^2 \\
&+ 2^{20} \sum_{i=0}^3 (a_{H,i}^{re,hi} + (b_i^{re,hl} - 8))^2 + 2^{16} \sum_{i=0}^3 ((a_{L,i}^{re,hi} - 8) + (b_i^{re,hl} - 8))^2 - 2^{20} \sum_{i=0}^3 (a_{H,i}^{im,hi} + (b_i^{im,hl} - 8))^2 - 2^{16} \sum_{i=0}^3 ((a_{L,i}^{im,hi} - 8) + (b_i^{im,hl} - 8))^2 \\
&+ 2^{12} \sum_{i=0}^3 ((a_{H,i}^{re,lo} - 8) + (b_i^{re,hl} - 8))^2 + 2^8 \sum_{i=0}^3 ((a_{L,i}^{re,lo} - 8) + (b_i^{re,hl} - 8))^2 - 2^{12} \sum_{i=0}^3 ((a_{H,i}^{im,lo} - 8) + (b_i^{im,hl} - 8))^2 - 2^8 \sum_{i=0}^3 ((a_{L,i}^{im,lo} - 8) + (b_i^{im,hl} - 8))^2 \\
&+ 2^{16} \sum_{i=0}^3 (a_{H,i}^{re,hi} + (b_i^{re,lh} - 8))^2 + 2^{12} \sum_{i=0}^3 ((a_{L,i}^{re,hi} - 8) + (b_i^{re,lh} - 8))^2 - 2^{16} \sum_{i=0}^3 (a_{H,i}^{im,hi} + (b_i^{im,lh} - 8))^2 - 2^{12} \sum_{i=0}^3 ((a_{L,i}^{im,hi} - 8) + (b_i^{im,lh} - 8))^2 \\
&+ 2^8 \sum_{i=0}^3 ((a_{H,i}^{re,lo} - 8) + (b_i^{re,lh} - 8))^2 + 2^4 \sum_{i=0}^3 ((a_{L,i}^{re,lo} - 8) + (b_i^{re,lh} - 8))^2 - 2^8 \sum_{i=0}^3 ((a_{H,i}^{im,lo} - 8) + (b_i^{im,lh} - 8))^2 - 2^4 \sum_{i=0}^3 ((a_{L,i}^{im,lo} - 8) + (b_i^{im,lh} - 8))^2 \\
&+ 2^{12} \sum_{i=0}^3 (a_{H,i}^{re,hi} + (b_i^{re,ll} - 8))^2 + 2^8 \sum_{i=0}^3 ((a_{L,i}^{re,hi} - 8) + (b_i^{re,ll} - 8))^2 - 2^{12} \sum_{i=0}^3 (a_{H,i}^{im,hi} + (b_i^{im,ll} - 8))^2 - 2^8 \sum_{i=0}^3 ((a_{L,i}^{im,hi} - 8) + (b_i^{im,ll} - 8))^2 \\
&+ 2^4 \sum_{i=0}^3 ((a_{H,i}^{re,lo} - 8) + (b_i^{re,ll} - 8))^2 + \sum_{i=0}^3 ((a_{L,i}^{re,lo} - 8) + (b_i^{re,ll} - 8))^2 - 2^4 \sum_{i=0}^3 ((a_{H,i}^{im,lo} - 8) + (b_i^{im,ll} - 8))^2 - \sum_{i=0}^3 ((a_{L,i}^{im,lo} - 8) + (b_i^{im,ll} - 8))^2 \\
\alpha &= 2^{24} \sum_{i=0}^3 (a_{H,i}^{re,hi})^2 + 2^{20} \sum_{i=0}^3 (a_{L,i}^{re,hi} - 8)^2 - 2^{24} \sum_{i=0}^3 (a_{H,i}^{im,hi})^2 - 2^{20} \sum_{i=0}^3 (a_{L,i}^{im,hi} - 8)^2 \\
&+ 2^{16} \sum_{i=0}^3 (a_{H,i}^{re,lo} - 8)^2 + 2^{12} \sum_{i=0}^3 (a_{L,i}^{re,lo} - 8)^2 - 2^{16} \sum_{i=0}^3 (a_{H,i}^{im,lo} - 8)^2 - 2^{12} \sum_{i=0}^3 (a_{L,i}^{im,lo} - 8)^2 \\
&+ 2^{20} \sum_{i=0}^3 (a_{H,i}^{re,hi} - 8)^2 + 2^{16} \sum_{i=0}^3 (a_{L,i}^{re,hi} - 16)^2 - 2^{20} \sum_{i=0}^3 (a_{H,i}^{im,hi} - 8)^2 - 2^{16} \sum_{i=0}^3 (a_{L,i}^{im,hi} - 16)^2 \\
&+ 2^{12} \sum_{i=0}^3 (a_{H,i}^{re,lo} - 16)^2 + 2^8 \sum_{i=0}^3 (a_{L,i}^{re,lo} - 16)^2 - 2^{12} \sum_{i=0}^3 (a_{H,i}^{im,lo} - 16)^2 - 2^8 \sum_{i=0}^3 (a_{L,i}^{im,lo} - 16)^2 \\
&+ 2^{16} \sum_{i=0}^3 (a_{H,i}^{re,hi} - 8)^2 + 2^{12} \sum_{i=0}^3 (a_{L,i}^{re,hi} - 16)^2 - 2^{16} \sum_{i=0}^3 (a_{H,i}^{im,hi} - 8)^2 - 2^{12} \sum_{i=0}^3 (a_{L,i}^{im,hi} - 16)^2 \\
&+ 2^8 \sum_{i=0}^3 (a_{H,i}^{re,lo} - 16)^2 + 2^4 \sum_{i=0}^3 (a_{L,i}^{re,lo} - 16)^2 - 2^8 \sum_{i=0}^3 (a_{H,i}^{im,lo} - 16)^2 - 2^4 \sum_{i=0}^3 (a_{L,i}^{im,lo} - 16)^2 \\
&+ 2^{12} \sum_{i=0}^3 (a_{H,i}^{re,hi} - 8)^2 + 2^8 \sum_{i=0}^3 (a_{L,i}^{re,hi} - 16)^2 - 2^{12} \sum_{i=0}^3 (a_{H,i}^{im,hi} - 8)^2 - 2^8 \sum_{i=0}^3 (a_{L,i}^{im,hi} - 16)^2 \\
&+ 2^4 \sum_{i=0}^3 (a_{H,i}^{re,lo} - 16)^2 + \sum_{i=0}^3 (a_{L,i}^{re,lo} - 16)^2 - 2^4 \sum_{i=0}^3 (a_{H,i}^{im,lo} - 16)^2 - \sum_{i=0}^3 (a_{L,i}^{im,lo} - 16)^2 \\
\beta &= 2^{24} \sum_{i=0}^3 (b_i^{re,hh})^2 + 2^{20} \sum_{i=0}^3 (b_i^{re,hh} - 8)^2 - 2^{24} \sum_{i=0}^3 (b_i^{im,hh})^2 - 2^{20} \sum_{i=0}^3 (b_i^{im,hh} - 8)^2 \\
&+ 2^{16} \sum_{i=0}^3 (b_i^{re,hh} - 8)^2 + 2^{12} \sum_{i=0}^3 (b_i^{re,hh} - 8)^2 - 2^{16} \sum_{i=0}^3 (b_i^{im,hh} - 8)^2 - 2^{12} \sum_{i=0}^3 (b_i^{im,hh} - 8)^2 \\
&+ 2^{20} \sum_{i=0}^3 (b_i^{re,hl} - 8)^2 + 2^{16} \sum_{i=0}^3 (b_i^{re,hl} - 16)^2 - 2^{20} \sum_{i=0}^3 (b_i^{im,hl} - 8)^2 - 2^{16} \sum_{i=0}^3 (b_i^{im,hl} - 16)^2 \\
&+ 2^{12} \sum_{i=0}^3 (b_i^{re,hl} - 16)^2 + 2^8 \sum_{i=0}^3 (b_i^{re,hl} - 16)^2 - 2^{12} \sum_{i=0}^3 (b_i^{im,hl} - 16)^2 - 2^8 \sum_{i=0}^3 (b_i^{im,hl} - 16)^2 \\
&+ 2^{16} \sum_{i=0}^3 (b_i^{re,lh} - 8)^2 + 2^{12} \sum_{i=0}^3 (b_i^{re,lh} - 16)^2 - 2^{16} \sum_{i=0}^3 (b_i^{im,lh} - 8)^2 - 2^{12} \sum_{i=0}^3 (b_i^{im,lh} - 16)^2 \\
&+ 2^8 \sum_{i=0}^3 (b_i^{re,lh} - 16)^2 + 2^4 \sum_{i=0}^3 (b_i^{re,lh} - 16)^2 - 2^8 \sum_{i=0}^3 (b_i^{im,lh} - 16)^2 - 2^4 \sum_{i=0}^3 (b_i^{im,lh} - 16)^2 \\
&+ 2^{12} \sum_{i=0}^3 (b_i^{re,ll} - 8)^2 + 2^8 \sum_{i=0}^3 (b_i^{re,ll} - 16)^2 - 2^{12} \sum_{i=0}^3 (b_i^{im,ll} - 8)^2 - 2^8 \sum_{i=0}^3 (b_i^{im,ll} - 16)^2 \\
&+ 2^4 \sum_{i=0}^3 (b_i^{re,ll} - 16)^2 + \sum_{i=0}^3 (b_i^{re,ll} - 16)^2 - 2^4 \sum_{i=0}^3 (b_i^{im,ll} - 16)^2 - \sum_{i=0}^3 (b_i^{im,ll} - 16)^2 \\
C &= 0 + 2^{20} \cdot 256 - 0 - 2^{20} \cdot 256 + 2^{16} \cdot 256 + 2^{12} \cdot 256 - 2^{16} \cdot 256 - 2^{12} \cdot 256 \\
&+ 2^{20} \cdot 256 + 2^{16} \cdot 1024 - 2^{20} \cdot 256 - 2^{16} \cdot 1024 + 2^{12} \cdot 1024 + 2^8 \cdot 1024 - 2^{12} \cdot 1024 - 2^8 \cdot 1024 \\
&+ 2^{16} \cdot 256 + 2^{12} \cdot 1024 - 2^{16} \cdot 256 - 2^{12} \cdot 1024 + 2^8 \cdot 1024 + 2^4 \cdot 1024 - 2^8 \cdot 1024 - 2^4 \cdot 1024 \\
&+ 2^{12} \cdot 256 + 2^8 \cdot 1024 - 2^{12} \cdot 256 - 2^8 \cdot 1024 + 2^4 \cdot 1024 + 1024 - 2^4 \cdot 1024 - 1024 = 0 \\
M^{re} &= \frac{1}{2} (PE - \alpha - \beta + C), \quad E = E_a + E_b
\end{aligned}$$

Im (M^{im}):

$$\begin{aligned}
PE &= 2^{24} \sum_{i=0}^3 (a_{H,i}^{re,hi} + b_i^{im,hh})^2 + 2^{20} \sum_{i=0}^3 ((a_{L,i}^{re,hi} - 8) + b_i^{im,hh})^2 + 2^{24} \sum_{i=0}^3 (a_{H,i}^{im,hi} + b_i^{re,hh})^2 + 2^{20} \sum_{i=0}^3 ((a_{L,i}^{im,hi} - 8) + b_i^{re,hh})^2 \\
&+ 2^{16} \sum_{i=0}^3 ((a_{H,i}^{re,lo} - 8) + b_i^{im,hh})^2 + 2^{12} \sum_{i=0}^3 ((a_{L,i}^{re,lo} - 8) + b_i^{im,hh})^2 + 2^{16} \sum_{i=0}^3 ((a_{H,i}^{im,lo} - 8) + b_i^{re,hh})^2 + 2^{12} \sum_{i=0}^3 ((a_{L,i}^{im,lo} - 8) + b_i^{re,hh})^2 \\
&+ 2^{20} \sum_{i=0}^3 (a_{H,i}^{re,hi} + (b_i^{im,hl} - 8))^2 + 2^{16} \sum_{i=0}^3 ((a_{L,i}^{re,hi} - 8) + (b_i^{im,hl} - 8))^2 + 2^{20} \sum_{i=0}^3 (a_{H,i}^{im,hi} + (b_i^{re,hl} - 8))^2 + 2^{16} \sum_{i=0}^3 ((a_{L,i}^{im,hi} - 8) + (b_i^{re,hl} - 8))^2 \\
&+ 2^{12} \sum_{i=0}^3 ((a_{H,i}^{re,lo} - 8) + (b_i^{im,hl} - 8))^2 + 2^8 \sum_{i=0}^3 ((a_{L,i}^{re,lo} - 8) + (b_i^{im,hl} - 8))^2 + 2^{12} \sum_{i=0}^3 ((a_{H,i}^{im,lo} - 8) + (b_i^{re,hl} - 8))^2 + 2^8 \sum_{i=0}^3 ((a_{L,i}^{im,lo} - 8) + (b_i^{re,hl} - 8))^2 \\
&+ 2^{16} \sum_{i=0}^3 (a_{H,i}^{re,hi} + (b_i^{im,hl} - 8))^2 + 2^{12} \sum_{i=0}^3 ((a_{L,i}^{re,hi} - 8) + (b_i^{im,hl} - 8))^2 + 2^{16} \sum_{i=0}^3 (a_{H,i}^{im,hi} + (b_i^{re,hl} - 8))^2 + 2^{12} \sum_{i=0}^3 ((a_{L,i}^{im,hi} - 8) + (b_i^{re,hl} - 8))^2 \\
&+ 2^8 \sum_{i=0}^3 ((a_{H,i}^{re,lo} - 8) + (b_i^{im,hl} - 8))^2 + 2^4 \sum_{i=0}^3 ((a_{L,i}^{re,lo} - 8) + (b_i^{im,hl} - 8))^2 + 2^8 \sum_{i=0}^3 ((a_{H,i}^{im,lo} - 8) + (b_i^{re,hl} - 8))^2 + 2^4 \sum_{i=0}^3 ((a_{L,i}^{im,lo} - 8) + (b_i^{re,hl} - 8))^2 \\
&+ 2^{12} \sum_{i=0}^3 (a_{H,i}^{re,hi} + (b_i^{im,ll} - 8))^2 + 2^8 \sum_{i=0}^3 ((a_{L,i}^{re,hi} - 8) + (b_i^{im,ll} - 8))^2 + 2^{12} \sum_{i=0}^3 (a_{H,i}^{im,hi} + (b_i^{re,ll} - 8))^2 + 2^8 \sum_{i=0}^3 ((a_{L,i}^{im,hi} - 8) + (b_i^{re,ll} - 8))^2 \\
&+ 2^4 \sum_{i=0}^3 ((a_{H,i}^{re,lo} - 8) + (b_i^{im,ll} - 8))^2 + \sum_{i=0}^3 ((a_{L,i}^{re,lo} - 8) + (b_i^{im,ll} - 8))^2 + 2^4 \sum_{i=0}^3 ((a_{H,i}^{im,lo} - 8) + (b_i^{re,ll} - 8))^2 + \sum_{i=0}^3 ((a_{L,i}^{im,lo} - 8) + (b_i^{re,ll} - 8))^2 \\
\alpha &= 2^{24} \sum_{i=0}^3 (a_{H,i}^{re,hi})^2 + 2^{20} \sum_{i=0}^3 (a_{L,i}^{re,hi} - 8)^2 + 2^{24} \sum_{i=0}^3 (a_{H,i}^{im,hi})^2 + 2^{20} \sum_{i=0}^3 (a_{L,i}^{im,hi} - 8)^2 \\
&+ 2^{16} \sum_{i=0}^3 (a_{H,i}^{re,lo} - 8)^2 + 2^{12} \sum_{i=0}^3 (a_{L,i}^{re,lo} - 8)^2 + 2^{16} \sum_{i=0}^3 (a_{H,i}^{im,lo} - 8)^2 + 2^{12} \sum_{i=0}^3 (a_{L,i}^{im,lo} - 8)^2 \\
&+ 2^{20} \sum_{i=0}^3 (a_{H,i}^{re,hi} - 8)^2 + 2^{16} \sum_{i=0}^3 (a_{L,i}^{re,hi} - 16)^2 + 2^{20} \sum_{i=0}^3 (a_{H,i}^{im,hi} - 8)^2 + 2^{16} \sum_{i=0}^3 (a_{L,i}^{im,hi} - 16)^2 \\
&+ 2^{12} \sum_{i=0}^3 (a_{H,i}^{re,lo} - 16)^2 + 2^8 \sum_{i=0}^3 (a_{L,i}^{re,lo} - 16)^2 + 2^{12} \sum_{i=0}^3 (a_{H,i}^{im,lo} - 16)^2 + 2^8 \sum_{i=0}^3 (a_{L,i}^{im,lo} - 16)^2 \\
&+ 2^{16} \sum_{i=0}^3 (a_{H,i}^{re,hi} - 8)^2 + 2^{12} \sum_{i=0}^3 (a_{L,i}^{re,hi} - 16)^2 + 2^{16} \sum_{i=0}^3 (a_{H,i}^{im,hi} - 8)^2 + 2^{12} \sum_{i=0}^3 (a_{L,i}^{im,hi} - 16)^2 \\
&+ 2^8 \sum_{i=0}^3 (a_{H,i}^{re,lo} - 16)^2 + 2^4 \sum_{i=0}^3 (a_{L,i}^{re,lo} - 16)^2 + 2^8 \sum_{i=0}^3 (a_{H,i}^{im,lo} - 16)^2 + 2^4 \sum_{i=0}^3 (a_{L,i}^{im,lo} - 16)^2 \\
&+ 2^{12} \sum_{i=0}^3 (a_{H,i}^{re,hi} - 8)^2 + 2^8 \sum_{i=0}^3 (a_{L,i}^{re,hi} - 16)^2 + 2^{12} \sum_{i=0}^3 (a_{H,i}^{im,hi} - 8)^2 + 2^8 \sum_{i=0}^3 (a_{L,i}^{im,hi} - 16)^2 \\
&+ 2^4 \sum_{i=0}^3 (a_{H,i}^{re,lo} - 16)^2 + \sum_{i=0}^3 (a_{L,i}^{re,lo} - 16)^2 + 2^4 \sum_{i=0}^3 (a_{H,i}^{im,lo} - 16)^2 + \sum_{i=0}^3 (a_{L,i}^{im,lo} - 16)^2 \\
\beta &= 2^{24} \sum_{i=0}^3 (b_i^{im,hh})^2 + 2^{20} \sum_{i=0}^3 (b_i^{im,hh} - 8)^2 + 2^{24} \sum_{i=0}^3 (b_i^{re,hh})^2 + 2^{20} \sum_{i=0}^3 (b_i^{re,hh} - 8)^2 \\
&+ 2^{16} \sum_{i=0}^3 (b_i^{im,hh} - 8)^2 + 2^{12} \sum_{i=0}^3 (b_i^{im,hh} - 8)^2 + 2^{16} \sum_{i=0}^3 (b_i^{re,hh} - 8)^2 + 2^{12} \sum_{i=0}^3 (b_i^{re,hh} - 8)^2 \\
&+ 2^{20} \sum_{i=0}^3 (b_i^{im,hl} - 8)^2 + 2^{16} \sum_{i=0}^3 (b_i^{im,hl} - 16)^2 + 2^{20} \sum_{i=0}^3 (b_i^{re,hl} - 8)^2 + 2^{16} \sum_{i=0}^3 (b_i^{re,hl} - 16)^2 \\
&+ 2^{12} \sum_{i=0}^3 (b_i^{im,hl} - 16)^2 + 2^8 \sum_{i=0}^3 (b_i^{im,hl} - 16)^2 + 2^{12} \sum_{i=0}^3 (b_i^{re,hl} - 16)^2 + 2^8 \sum_{i=0}^3 (b_i^{re,hl} - 16)^2 \\
&+ 2^{16} \sum_{i=0}^3 (b_i^{im,hl} - 8)^2 + 2^{12} \sum_{i=0}^3 (b_i^{im,hl} - 16)^2 + 2^{16} \sum_{i=0}^3 (b_i^{re,hl} - 8)^2 + 2^{12} \sum_{i=0}^3 (b_i^{re,hl} - 16)^2 \\
&+ 2^8 \sum_{i=0}^3 (b_i^{im,hl} - 16)^2 + 2^4 \sum_{i=0}^3 (b_i^{im,hl} - 16)^2 + 2^8 \sum_{i=0}^3 (b_i^{re,hl} - 16)^2 + 2^4 \sum_{i=0}^3 (b_i^{re,hl} - 16)^2 \\
&+ 2^{12} \sum_{i=0}^3 (b_i^{im,ll} - 8)^2 + 2^8 \sum_{i=0}^3 (b_i^{im,ll} - 16)^2 + 2^{12} \sum_{i=0}^3 (b_i^{re,ll} - 8)^2 + 2^8 \sum_{i=0}^3 (b_i^{re,ll} - 16)^2 \\
&+ 2^4 \sum_{i=0}^3 (b_i^{im,ll} - 16)^2 + \sum_{i=0}^3 (b_i^{im,ll} - 16)^2 + 2^4 \sum_{i=0}^3 (b_i^{re,ll} - 16)^2 + \sum_{i=0}^3 (b_i^{re,ll} - 16)^2 \\
C &= 0 + 2^{20} \cdot 256 + 0 + 2^{20} \cdot 256 + 2^{16} \cdot 256 + 2^{12} \cdot 256 + 2^{16} \cdot 256 + 2^{12} \cdot 256 \\
&+ 2^{20} \cdot 256 + 2^{16} \cdot 1024 + 2^{20} \cdot 256 + 2^{16} \cdot 1024 + 2^{12} \cdot 1024 + 2^8 \cdot 1024 + 2^{12} \cdot 1024 + 2^8 \cdot 1024 \\
&+ 2^{16} \cdot 256 + 2^{12} \cdot 1024 + 2^{16} \cdot 256 + 2^{12} \cdot 1024 + 2^8 \cdot 1024 + 2^4 \cdot 1024 + 2^8 \cdot 1024 + 2^4 \cdot 1024 \\
&+ 2^{12} \cdot 256 + 2^8 \cdot 1024 + 2^{12} \cdot 256 + 2^8 \cdot 1024 + 2^4 \cdot 1024 + 1024 + 2^4 \cdot 1024 + 1024 = 1297680384 \\
M^{im} &= \frac{1}{2} (PE - \alpha - \beta + C), \quad E = E_a + E_b
\end{aligned}$$